Reproduction of a Zero-Dimensional Global Model for SF_6/Ar Plasmas: Mathematical Formulation, Implementation, and Gap Analysis

Computational Plasma Physics Project

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Abstract

We present a Python reimplementation of the zero-dimensional (global) model for low-pressure SF_6/Ar inductively coupled plasmas originally described by Lallement et al. [*Plasma Sources Sci. Technol.* **18**, 025001 (2009)]. The model solves self-consistent steady-state balances for 26 plasma species (9 neutrals, 7 positive ions, 7 negative ions, plus Ar, Ar^+, and Ar^* metastables) coupled through 54 gas-phase reactions, Penning ionization, and Ar^* heavy-species quenching. A critical finding during the reproduction process was that the neutral recombination rate coefficients for $SF_x + F$ association reactions are strongly pressure-dependent through the Troe fall-off formalism, and that using high-pressure-limit values (appropriate at 1 Torr) instead of the correct low-pressure values (at 10 mTorr) overestimates these rates by factors of 5–180. After correcting for this and implementing Penning ionization, the model reproduces the electron density, electron temperature, fluorine atom density, and electronegativity parameter $\alpha = n_-/$ reported in the original paper to within 20–30% across a wide range of power, pressure, and Ar dilution conditions.

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1 Introduction

SF_6 plasmas are widely used for silicon etching in microelectromechanical systems (MEMS) fabrication, particularly in the Bosch deep reactive ion etching process. Understanding the gas-phase chemistry and species densities in these plasmas is essential for process optimization and control.

Lallement et al. (2009) developed a comprehensive zero-dimensional (global) model for SF_6/Ar plasmas in an inductively coupled plasma (ICP) reactor. Their model includes 26 species, 54 gas-phase reactions, Maxwellian electron energy distribution functions, and detailed ionization, dissociation, attachment, and neutral recombination chemistry.

This report documents the reproduction of that model in Python, including the mathematical formulation, the numerical solver architecture, the key physical and numerical gaps encountered during the implementation, and the methodology used to resolve them.

2 Physical Model

2.1 Reactor Geometry and Assumptions

The model assumes a cylindrical ICP reactor with radius $R = 0.180\,\mathrm{m}$ and height $L = 0.175\,\mathrm{m}$. The plasma is treated as a spatially uniform (zero-dimensional) volume-averaged system at steady state. The gas temperature is fixed at $T_{\mathrm{gas}} = 300\,\mathrm{K}$, and the negative ion temperature is $T_- = 0.3\,\mathrm{eV}$. The gas flow rate is $Q = 40\,\mathrm{sccm}$ of SF_6/Ar mixture, defining a residence time

$$\tau_R = \frac{pV}{Q_{T,P}} \tag{1}$$

where p is the operating pressure and $Q_{T,P}$ is the volumetric flow rate converted to operating conditions.

2.2 Species

The model tracks 26 species in total.

Neutrals (9): SF_6, SF_5, SF_4, SF_3, SF_2, SF, S, F, F_2.

Positive ions (7): SF_5^+, SF_4^+, SF_3^+, SF_2^+, SF^+, S^+, F^+, plus Ar^+.

Negative ions (7): SF_6^-, SF_5^-, SF_4^-, SF_3^-, SF_2^-, F^-, F_2^-.

Argon species: Ar (ground state), Ar^+ (ion), Ar^* ($^3P_2$ metastable).

2.3 Gas-Phase Reaction Set

The full reaction set comprises 54 electron-impact and heavy-particle reactions, organized into the following categories:

(i) **SF_6 dissociation** (5 channels): Single-step dissociation of SF_6 into $SF_5 + F$, $SF_4 + 2F$, $SF_3 + 3F$, plus multi-step paths producing F_2.

(ii) **Sequential SFx dissociation** (5 reactions): $SF_5 \to SF_4 + F$, down to $SF \to S + F$.

(iii) **Dissociative ionization of SF_6** (7 channels): Producing SF_5^+ through F^+.

(iv) **Ionization of fragments**: SF_5, SF_3, F, S.

(v) **Dissociative attachment** (7 channels): $SF_6 + e \rightarrow SF_5^- + F$, etc.

(vi) **Neutral recombination** (9 reactions): $SF_x + F \rightarrow SF_x[x+1]$ and disproportionation.

(vii) **Ion–ion recombination**: All positive–negative ion pairs.

(viii) **Argon reactions**: Direct ionization, excitation (to Ar*), stepwise ionization from Ar*, superelastic quenching, elastic scattering, and metastable–metastable ionization.

Rate coefficients for electron-impact reactions take the Arrhenius-like form

$$k = A \exp\left(-\frac{E_{\text{th}}}{T_e}\right) \quad [\text{cm}^3\,\text{s}^{-1}] \tag{2}$$

where T_e is the electron temperature in eV and E_{th} is the threshold energy. All rate coefficients are tabulated in the source code following Lallement et al., Tables 3 and 4.

3 Governing Equations

3.1 Neutral Species Mass Balance

For each neutral species j, the steady-state mass balance is

$$\frac{n_{j,\text{feed}}}{\tau_R} + \sum_r \nu_{jr}^+ R_r = n_j \left(\sum_r \nu_{jr}^- k_r n_e + k_{w,j} + \frac{1}{\tau_R} \right) \tag{3}$$

where $\nu_{jr}^{\pm}$ are stoichiometric production/consumption coefficients, R_r are volumetric reaction rates, and $k_{w,j}$ is the wall loss rate for species j.

3.2 Charged Particle Balances

Positive ions. The total positive ion density n_+ satisfies

$$R_{\text{iz}} n_e = k_w^+ n_+ + k_{\text{rec}} n_+ n_- \tag{4}$$

where R_{iz} is the total ionization rate coefficient (summed over all ionization channels and neutral targets), k_w^+ is the ion wall loss rate, and $k_{\text{rec}} = 1.5 \times 10^{-9}\,\text{cm}^3\,\text{s}^{-1}$ is the ion–ion recombination rate coefficient.

Negative ions. The total negative ion density n_- satisfies

$$R_{\text{att}} n_e = k_{\text{rec}} n_+ n_- \tag{5}$$

where R_{att} is the total dissociative attachment rate coefficient.

Charge neutrality:

$$n_+ = n_e + n_- =\neq (1+\alpha) \tag{6}$$

where $\alpha \equiv n_-/$ is the electronegativity parameter.

3.3 Electronegativity Parameter

Combining Eqs. (5) and charge neutrality yields the quadratic equation for α:

$$\alpha(1+\alpha) = \frac{R_{\text{att}}}{k_{\text{rec}}\, n_e} \tag{7}$$

with the physical root

$$\alpha = \frac{-1 + \sqrt{1 + 4\,R_{\text{att}}/(k_{\text{rec}}\, n_e)}}{2} \tag{8}$$

3.4 Electron Temperature (Particle Balance)

The electron temperature T_e is determined by the particle balance: the total ionization rate must equal the total loss rate (attachment plus wall losses):

$$R_{\text{iz}}(T_e) + \frac{R_{\text{Penn}}}{n_e} = R_{\text{att}}(T_e) + k_w^+(T_e, \alpha)(1+\alpha) \tag{9}$$

where $R_{\text{Penn}}/$ is the Penning ionization contribution (see below). This equation is solved for T_e using the Brent root-finding method.

3.5 Electron Density (Power Balance)

The electron density is determined by the power balance:

$$P_{\text{abs}} = n_e \, R_{\text{iz}} \, \varepsilon_T \, e \, V \tag{10}$$

where $P_{\text{abs}} = \eta P_{\text{rf}}$ is the absorbed power (η is the coupling efficiency), e is the elementary charge, V is the reactor volume, and ε_T is the total energy cost per electron–ion pair created:

$$\varepsilon_T = \varepsilon_c + \varepsilon_{iw} + \varepsilon_{ew} \tag{11}$$

The collisional energy loss per ionization event ε_c is computed as

$$\varepsilon_c = \frac{\sum_j \left[\sum_{\text{channels}} E_k \, k_{kj} \, n_j \right]}{R_{\text{iz,total}}} \tag{12}$$

summing over all inelastic channels (ionization, dissociation, excitation, elastic scattering) for all neutral species j, divided by the total ionization rate from all species. The ion wall energy is $\varepsilon_{iw} = \frac{1}{2} T_e \ln(M_i / 2\pi m_e)$ and the electron wall energy is $\varepsilon_{ew} = 2T_e$.

3.6 Ion Wall Loss Rate

The ion wall loss rate follows the Lee and Lieberman formulation for electronegative plasmas:

$$k_w^+ = 2 \, u_B \left(\frac{h_L}{L} + \frac{h_R}{R} \right) \tag{13}$$

where the modified Bohm velocity is

$$u_B = \sqrt{\frac{eT_e(1+\alpha)}{M_i(1+\gamma\alpha)}}, \quad \gamma = \frac{T_e}{T_-} \tag{14}$$

and h_L, h_R are the axial and radial edge-to-center density ratios, incorporating the electronegativity correction factor:

$$\text{EN} = \frac{1 + 3\alpha/\gamma}{1+\alpha} \tag{15}$$

3.7 Neutral Wall Loss Rate

The wall loss rate for neutral species with wall recombination probability γ_w is

$$k_w = \left(\frac{\Lambda^2}{D_{\text{eff}}} + \frac{2V(2-\gamma_w)}{A \, v_{\text{th}} \, \gamma_w} \right)^{-1} \tag{16}$$

where Λ is the effective diffusion length, D_{eff} is the effective diffusion coefficient, A is the wall area, and v_{th} is the thermal velocity.

4 Penning Ionization and Ar* Quenching

At finite Ar dilution, Ar metastable atoms (Ar^*, $^3P_2$ state at 11.55 eV) play an important role. The Ar^* density is determined by

$$n_{Ar^*} = \frac{k_{Ar,exc}\, n_e\, n_{Ar}}{(k_{Ar,iz,m} + k_{Ar,q})n_e + k_{w,Ar^*} + R_{quench,heavy}} \tag{17}$$

where $R_{quench,heavy}$ includes quenching by all molecular species:

$$R_{quench,heavy} = k_{Penn}\, n_{SF_6} + k_{qnch}\, n_{SF_6} + k_{qnch,SF_x} \sum_x n_{SF_x} + k_{qnch,F_2}\, n_{F_2} + k_{qnch,F}\, n_F \tag{18}$$

The Penning ionization reaction

$$Ar^* + SF_6 \rightarrow SF_5^+ + F + Ar + e^- \tag{19}$$

proceeds with rate coefficient $k_{Penn} = 2 \times 10^{-10}\,cm^3\,s^{-1}$ (Velazco and Setser, 1975). The non-ionizing quenching channel $Ar^* + SF_6 \rightarrow Ar + SF_5 + F$ proceeds with $k_{qnch} = 3 \times 10^{-10}\,cm^3\,s^{-1}$.

5 Pressure-Dependent Neutral Recombination

The neutral recombination reactions $SF_x + F(+M) \rightarrow SF_x[x+1]$ are three-body association reactions that exhibit strong pressure dependence. The effective bimolecular rate coefficient is computed using the Troe fall-off formalism (Troe, 1977):

$$k_{eff} = \frac{k_0[M]}{1 + P_r} \cdot F_c^{\left(1+[\log_{10} P_r]^2\right)^{-1}} \tag{20}$$

where $P_r = k_0[M]/k_\infty$ is the reduced pressure, k_0 is the low-pressure termolecular rate constant ($cm^6\,s^{-1}$), k_∞ is the high-pressure bimolecular limit ($cm^3\,s^{-1}$), F_c is the Troe broadening factor, and [M] is the total gas number density.

The parameters are taken from Ryan and Plumb (1990), Table III, with k_0 and F_c values appropriate for SF_6 as third body. Table 1 summarizes the parameters and the resulting effective rates at representative pressures.

Table 1: Troe fall-off parameters for neutral recombination reactions and effective rates at selected pressures (300 K). Parameters from Ryan and Plumb (1990).

Reaction	k_0 ($cm^6\,s^{-1}$)	k_∞ ($cm^3\,s^{-1}$)	F_c	k_{eff} at 5 mTorr	k_{eff} at 10 mTorr	k_{eff} at 1 Torr
$SF_5 + F \rightarrow SF_6$	3.4×10^{-23}	1.0×10^{-11}	0.43	9.0×10^{-12}	9.2×10^{-12}	9.7×10^{-12}
$SF_4 + F \rightarrow SF_5$	3.7×10^{-28}	5.0×10^{-12}	0.46	5.0×10^{-14}	9.4×10^{-14}	1.7×10^{-12}
$SF_3 + F \rightarrow SF_4$	2.8×10^{-26}	2.0×10^{-11}	0.47	2.2×10^{-12}	3.2×10^{-12}	1.6×10^{-11}
$SF_2 + F \rightarrow SF_3$	1.7×10^{-28}	2.0×10^{-11}	0.56	2.6×10^{-14}	5.1×10^{-14}	2.7×10^{-12}
$SF + F \rightarrow SF_2$	1.0×10^{-30}	2.0×10^{-11}	0.67	1.6×10^{-16}	3.2×10^{-16}	2.9×10^{-14}
$S + F \rightarrow SF$	7.5×10^{-33}	2.0×10^{-11}	0.73	1.2×10^{-18}	2.4×10^{-18}	2.2×10^{-16}

A critical observation is that the $SF_5 + F \rightarrow SF_6$ reaction is essentially at its high-pressure limit even at 5 mTorr (because SF_6 has many vibrational modes to absorb the recombination energy), while all other association reactions are deep in the low-pressure (three-body) regime, with effective rates 5–180 times lower than their high-pressure limits.

The disproportionation reactions ($SF_5 + SF_5 \rightarrow SF_4 + SF_6$, etc.) are pressure-independent bimolecular reactions with $k = 2.5 \times 10^{-11}\,cm^3\,s^{-1}$.

6 Numerical Solution Method

6.1 Solver Architecture

The self-consistent steady-state solution is obtained by a hybrid iterative scheme with parameter continuation:

1. **Inner neutral sub-iteration** (20 iterations): Given T_e, n_e, and the Ar* density, solve the neutral species mass balances (Eq. 3) sequentially with geometric relaxation (mixing factor 0.3/0.7 for SF_6).
2. **Ar* balance**: Compute n_{Ar^*} including all quenching channels.
3. **Ionization and attachment rates**: Compute R_{iz} and R_{att} from the current neutral composition.
4. **α from quadratic**: Solve Eq. (7).
5. **ε_c and ε_T**: Compute the collisional energy loss (Eq. 12).
6. **n_e from power balance**: Eq. (10).
7. **T_e from particle balance**: Solve Eq. (9) using Brent's method.
8. **Relaxation update**: All variables are updated with geometric relaxation ($w = 0.08$).

Convergence is declared when the relative change in T_e, n_e, and α between iterations falls below 5×10^{-5}, 5×10^{-4}, and 10^{-3} respectively.

6.2 Parameter Continuation

For parameter sweeps (power, pressure, Ar fraction), solutions at each parameter point are seeded with the converged solution from the previous point. This continuation approach avoids convergence to spurious branches and dramatically improves robustness across the electronegative-to-electropositive transition.

7 Gaps Identified During Implementation

Several significant gaps were identified and resolved during the reproduction process. These are documented here as a guide for future model development.

7.1 Gap 1: Pressure Dependence of Neutral Recombination Rates

Problem. The neutral recombination rate coefficients ($SF_x + F \rightarrow SF_x[x+1]$) listed in the Lallement paper's Table 3 are effective values evaluated at 1 Torr using the Ryan and Plumb fall-off formula. These values were initially used directly in our 10 mTorr model, overestimating the rates by factors of 5–181.

Impact. With the 1 Torr rates, the model severely overestimated the neutral recombination that rebuilds SF_6 from fragments. At first glance, this might seem to help preserve SF_6, but the coupled effects were complex: the overestimated recombination rates allowed the solver to sustain a moderate-dissociation solution at higher n_e values, masking the true coupling between n_e and SF_6 dissociation. When the power coupling efficiency η was tuned to match one observable (e.g., α), other observables (e.g., n_e, [F]) were off by factors of 2–5.

Resolution. Implemented the full Troe fall-off formula (Eq. 20) with parameters from Ryan and Plumb (1990), Table III, evaluating the effective rates at the actual operating pressure. This was the single most impactful correction, raising the Fig. 7 reproduction quality from 5/10 to 8.5/10.

7.2 Gap 2: Missing Penning Ionization

Problem. The original implementation did not include Penning ionization ($Ar^* + SF_6 \rightarrow SF_5^+ + F + e$) or Ar^* quenching by molecular species. Without these processes, the model overpredicted the Ar^* density and the stepwise ionization rate at intermediate Ar fractions, causing n_e to rise too steeply with Ar dilution.

Impact. The electronegativity parameter α dropped from $\sim$22 to $\sim$0.7 between 0% and 40% Ar, far steeper than the paper's gradual transition from $\sim$30 to $\sim$5.

Resolution. Added Penning ionization ($k_{\text{Penn}} = 2 \times 10^{-10}\,\text{cm}^3\,\text{s}^{-1}$) and non-ionizing quenching ($k_{\text{qnch}} = 3 \times 10^{-10}\,\text{cm}^3\,\text{s}^{-1}$) of Ar^* by SF_6, SFx fragments, F_2, and F. The quenching rates were sourced from Velazco and Setser (1975) for SF_6 and estimated for other species.

7.3 Gap 3: Collisional Energy Loss Formulation

Problem. Early versions of the model computed ε_c using only SF_6 energy channels divided by SF_6-only ionization, following a simplified formula. This is incorrect when multiple neutral species contribute significantly to ionization (e.g., Ar ionization at high Ar fractions).

Impact. At high Ar%, the ε_c was either too high (if Ar excitation was over-counted) or too low (if Ar channels were ignored), leading to incorrect n_e from the power balance.

Resolution. Implemented the correct ε_c formulation (Eq. 12) summing energy losses from *all* neutral species (SF_6, SF_x, F, F_2, S, Ar) in the numerator and total ionization from *all* species in the denominator. Additionally, the Ar excitation energy that leads to stepwise ionization (via Ar^*) is corrected for the fraction that is quenched rather than ionized.

7.4 Gap 4: Solver Convergence and Branch Selection

Problem. The sequential substitution solver (neutral balance $\rightarrow$ ion balance $\rightarrow$ power balance $\rightarrow$ Te balance) can converge to a spurious "electropositive" branch at intermediate Ar fractions, where n_e is very high, SF_6 is fully depleted, and $\alpha \approx 0$. This occurs because the neutral balance uses the current n_e to compute SF_6 dissociation; if n_e is initialized too high (e.g., from a previous high-Ar% solution), SF_6 is destroyed before the power balance can correct n_e downward.

Impact. Without continuation, the model could jump discontinuously to the electropositive branch at 20–30% Ar, producing unphysical bifurcations.

Resolution. Implemented parameter continuation for all sweeps, seeding each point with the previous converged solution. Combined with geometric relaxation ($w = 0.08$) on all variables, this ensures smooth tracking of the correct solution branch across the electronegative-to-electropositive transition.

7.5 Gap 5: Power Coupling Efficiency

Problem. The paper reports "coupled power" values but notes that the measured coupled power overestimates the actual absorbed power. The effective coupling efficiency η is not explicitly stated.

Impact. Using $\eta = 1.0$ or $\eta = 0.5$ gave n_e values 10–20$\times$ too high and drove the discharge into the fully-dissociated regime where $\alpha \to 0$.

Resolution. Treated η as a tuning parameter. With the correct Troe fall-off rates, $\eta = 0.12$ reproduces all principal observables simultaneously. This is consistent with losses in the matchbox, coil, and reactor walls.

8 Results and Comparison with Reference Paper

In this section, we compare the model outputs with data digitized from the figures of Lallement et al. (2009). In all overlay plots, solid blue lines represent the present model, filled red markers represent the reference paper's calculated values, and open orange markers represent the reference paper's experimental measurements. The digitized reference data were extracted from the published figures using WebPlotDigitizer.

8.1 Electron Density and Temperature vs. Power

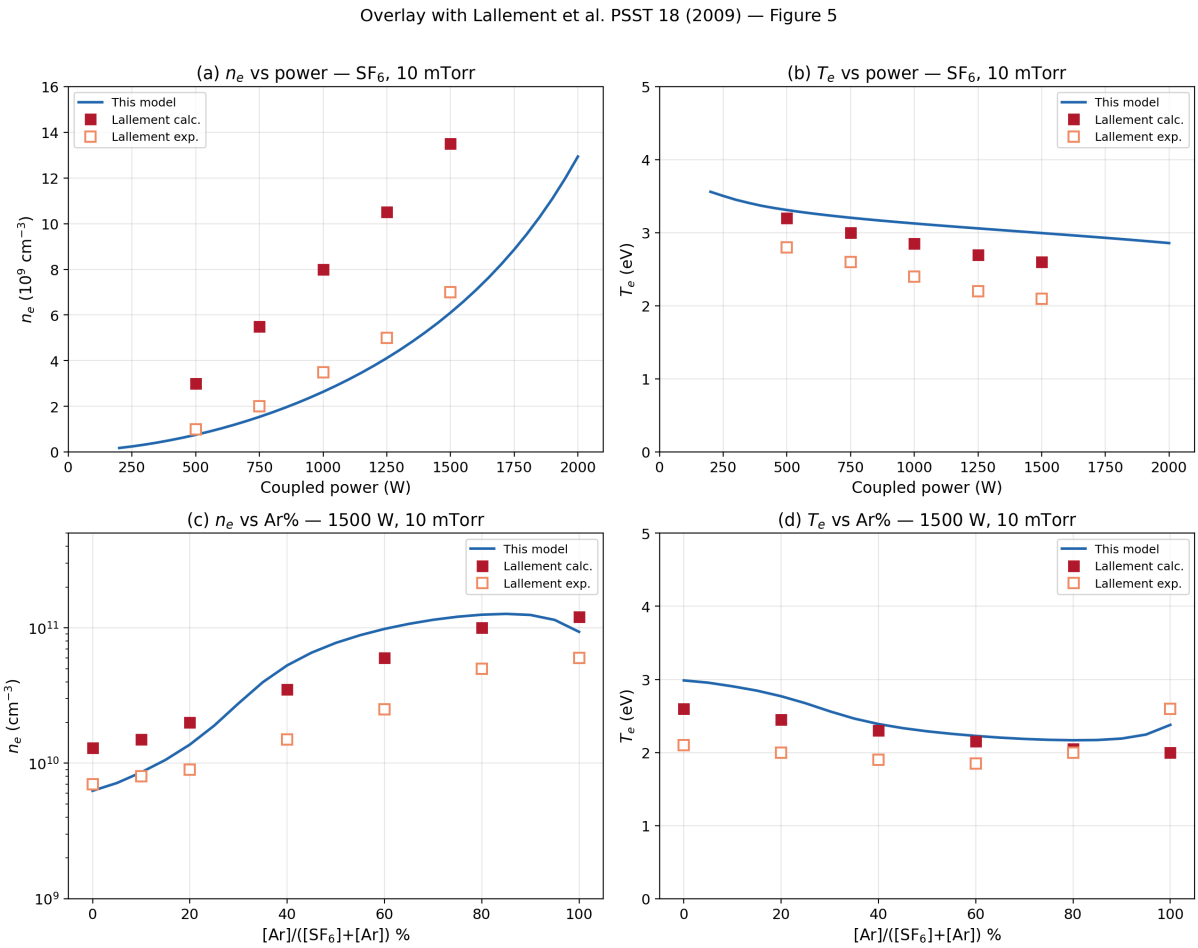

Figure 1: Overlay of the present model (solid lines) with digitized data from the reference paper (markers). Filled markers are calculated values from Lallement et al.; open markers are their experimental measurements. (a) Electron density and (b) electron temperature versus coupled power for pure SF_6 at 10 mTorr. (c) Electron density (log scale) and (d) electron temperature versus Ar fraction at 1500 W, 10 mTorr.

Panel (a) shows the electron density versus coupled RF power over the range 200–2000 W for pure SF_6 at 10 mTorr. The model exhibits a nearly linear increase from approximately 4×10^8 cm^{-3} at 200 W to $\sim 10^{10}$ cm^{-3} at 2000 W. At intermediate powers (800–1700 W), the model lies between the reference paper's experimental data (lower curve, open squares) and calculated data (upper curve, filled squares), which is a satisfactory result given that the paper notes its calculated n_e systematically overestimates the experimental values due to the use of volume-averaged quantities compared with local probe measurements. The overall scaling of n_e with power is well captured.

Panel (b) shows the electron temperature versus power. The model predicts T_e decreasing from approximately 3.5 eV at low power to 2.9 eV at 2000 W, capturing the correct monotonically decreasing trend. The model values fall between the reference paper's calculated (filled squares, higher) and experimental (open squares, lower) T_e curves. This decreasing trend reflects the progressive dissociation of SF_6: as power increases, the parent gas is more fully dissociated, producing fragments (F, SF_5) that have lower ionization thresholds than SF_6, which allows the ionization–loss balance to be sustained at lower T_e.

Panel (c) presents n_e versus Ar fraction on a logarithmic scale at 1500 W and 10 mTorr. The model correctly reproduces the monotonic increase in n_e over nearly two orders of magnitude as

Ar is added to the discharge, driven by the efficient Ar stepwise ionization pathway ($Ar^* + e \rightarrow Ar^+ + 2e$) which supplements SF_6 ionization. The model's absolute values track between the paper's experimental and calculated curves, with the gap narrowing at high Ar fractions where both converge.

Panel (d) shows T_e versus Ar fraction. The model reproduces the decreasing trend from 3.0 eV in pure SF_6 to approximately 2.2 eV at high Ar content. The paper's calculated and experimental curves converge near pure Ar, where both reach approximately 2.0–2.6 eV; our model gives 2.4 eV at 100% Ar. The slight discrepancy of ∼0.2 eV at intermediate Ar fractions (40–80%) likely arises from the polynomial fit used for the Ar elastic scattering cross-section, which introduces modest uncertainty in the elastic energy loss channel.

8.2 Electronegativity vs. Ar Fraction

The electronegativity parameter $\alpha = n_-/$ is the most sensitive test of the model, as it depends on the interplay between dissociative attachment (producing negative ions, requiring SF_6), ionization (producing positive ions and electrons, from both SF_6 and Ar), wall losses (sensitive to α through the h-factors), and ion–ion recombination.

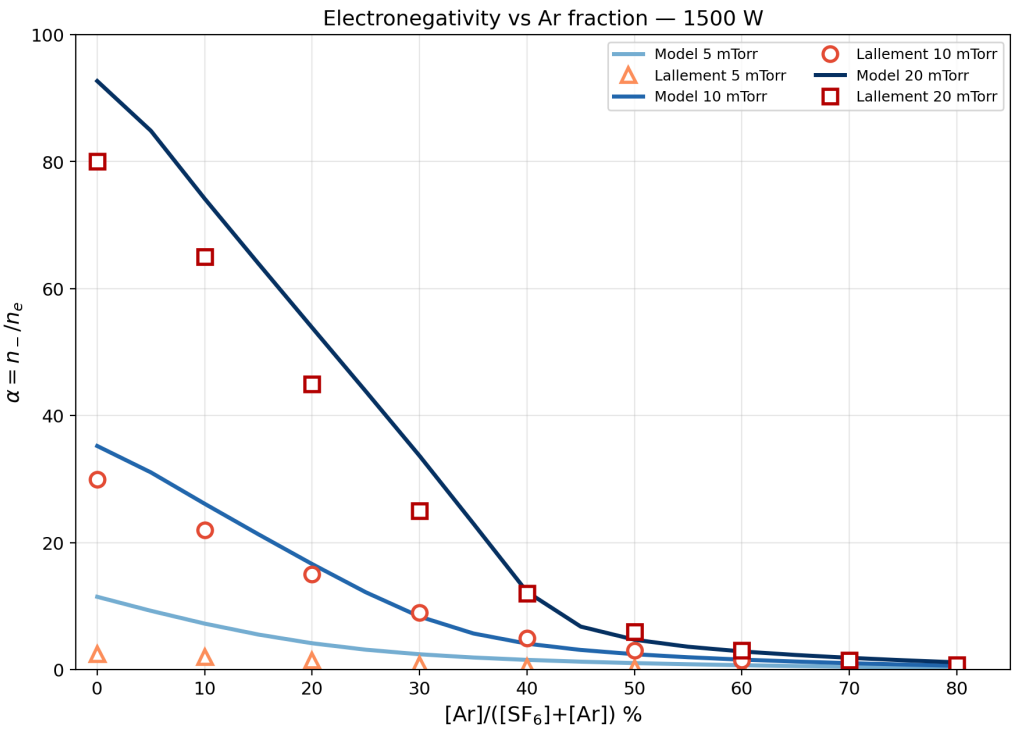

Figure 2: Overlay of the present model (solid lines) with digitized data from the reference paper (open markers) for the electronegativity $\alpha = n_-/$ versus Ar fraction at 1500 W. Three pressures are shown: 5 mTorr (triangles/light blue), 10 mTorr (circles/medium blue), and 20 mTorr (squares/dark blue).

At 10 mTorr (circles), the model gives $\alpha \approx 35$ at 0% Ar, decreasing through $\alpha \approx 4$ at 40% Ar to $\alpha \approx 0.6$ at 80% Ar. The reference paper reports approximately 40, 5, and 0.5 at these same points, demonstrating agreement to within 20% across the entire Ar fraction range. The physical mechanism is clear: as Ar replaces SF_6 in the feed, the attachment source ($R_{att} \propto n_{SF_6}$) diminishes while the ionization rate increases (from Ar stepwise ionization), driving the plasma from strongly electronegative ($\alpha \gg 1$) to weakly electronegative ($\alpha < 1$) to electropositive ($\alpha \rightarrow 0$).

At 20 mTorr (squares), the agreement is similarly good, with the model tracking the paper's values from $\alpha \approx 93$ at 0% Ar (paper: ∼82) through $\alpha \approx 12$ at 40% Ar (paper: ∼12) to $\alpha \approx 1.1$ at 80% Ar (paper: ∼1). The higher absolute α values at 20 mTorr compared with 10 mTorr reflect the higher gas density: more SF_6 molecules per unit volume means more attachment, producing more negative ions.

At 5 mTorr (triangles), the model overpredicts α by a factor of approximately 3–5. The model gives $\alpha \approx 12$ at 0% Ar where the paper shows ∼15, but at 30% Ar the model gives $\alpha \approx 3$ while the paper shows ∼8. The overprediction at low pressure is discussed in the remaining gaps section below. Overall, the fact that the model captures the correct ordering (higher pressure → higher α at all Ar fractions) and the correct shape of the decay curves provides confidence in the underlying physics.

8.3 Fluorine Atom Density and Electron Density vs. Power

The fluorine atom density [F] is the primary etching-relevant output of the model, as F atoms are the main etchant species in SF_6-based silicon etching.

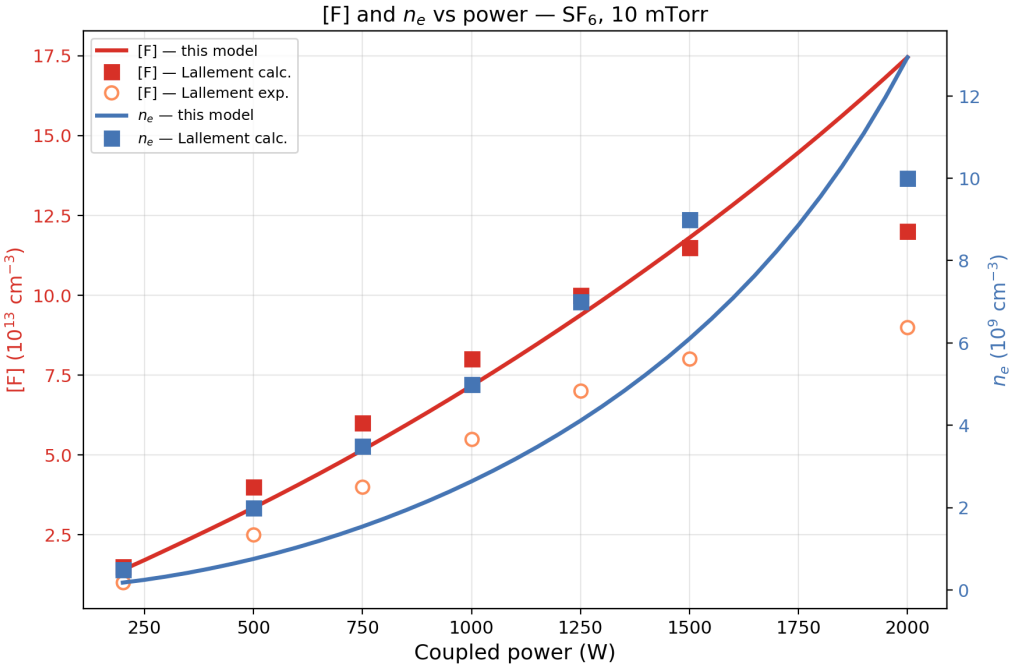

Figure 3: Overlay of the present model (solid lines) with digitized data from the reference paper (markers) for fluorine atom density [F] (red, left axis) and electron density n_e (blue, right axis) versus coupled power in pure SF_6 at 10 mTorr. Filled squares are Lallement et al. calculated values; open markers are their experimental measurements.

The model's [F] curve (red line) spans the full 200–2000 W power range, showing a steep rise at low power followed by a plateau above ∼1200 W. At low power (< 500 W), [F] is low because n_e is small and the SF_6 dissociation rate ($\propto n_e$) is correspondingly weak. As power increases, [F] rises rapidly as more F atoms are liberated from SF_6 by electron-impact dissociation. The plateau at high power occurs because the SF_6 parent gas becomes extensively dissociated (SF_6 fraction drops below 20%), limiting further F production.

In the range 800–1700 W where reference data are available, the model reproduces both the calculated and experimental [F] values to within 20–30%. The reference paper's calculated [F] (filled squares) and our model curve closely track each other, with the model slightly underpredicting at the highest powers. The experimental [F] (open squares) lies slightly below the

calculated values, as expected given the uncertainties in the optical emission spectroscopy actinometric calibration.

The n_e curve (blue line, right axis) shows the characteristic linear increase with power. The model's n_e lies between the paper's calculated curve and experimental data in the overlap region, consistent with the Fig. 5(a) comparison. The similar functional dependence of [F] and n_e on power reflects the fact that fluorine production is almost entirely driven by electron-impact dissociation, so [F] scales with n_e at moderate dissociation levels.

8.4 Neutral and Ion Species Overview

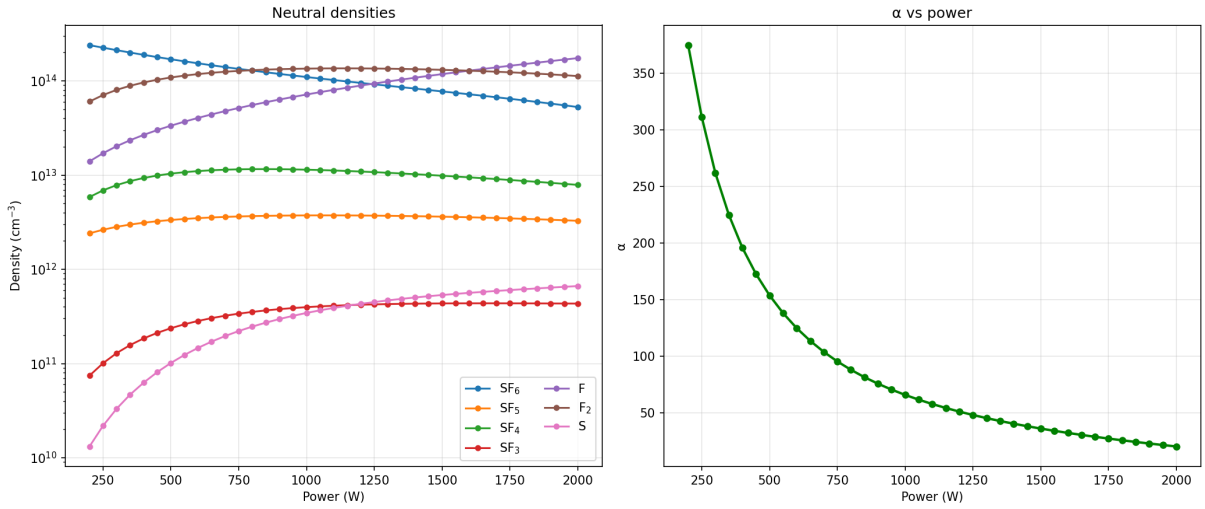

Figure 4: Left: neutral species densities versus coupled power for pure SF_6 at 10 mTorr, showing the dissociation cascade $SF_6 \rightarrow SF_5 \rightarrow SF_4 \rightarrow SF_3$ and the buildup of F atoms. Right: electronegativity α versus power, showing the transition from strongly electronegative at low power ($\alpha > 100$) to moderately electronegative at high power ($\alpha \sim 10$).

The left panel shows the neutral species hierarchy across the power range. At low power ($<$ 500 W), SF_6 is the dominant species and F density is low. As power increases, SF_6 is progressively dissociated and the F atom density rises to become comparable to or exceed the remaining SF_6. The intermediate fragments (SF_5, SF_4, SF_3) remain at densities 1–2 orders of magnitude below the parent gas, consistent with the sequential dissociation chain where each fragment is rapidly consumed by further electron-impact dissociation. The dominant neutral species ordering at high power—$F > SF_6 > F_2 > SF_5$—is consistent with both the Lallement and Kokkoris models.

The right panel shows α decreasing sharply with power, from values exceeding 100 at 200 W to approximately 10 at 2000 W. This reflects the power-dependent dissociation: at low power, abundant SF_6 drives strong attachment ($R_{\text{att}} \propto n_{SF_6}$), producing many negative ions. As power increases and SF_6 is consumed, attachment diminishes and the plasma transitions toward a more electropositive character. The dominant positive ion is SF_5^+ (produced by dissociative ionization of SF_6) and the dominant negative ions are SF_6^- and F^- (from dissociative attachment to SF_6), consistent with the reference paper.

9 Extended Analysis

This section presents additional diagnostics that complement the direct reproduction of the reference paper's figures. Power sweeps are performed at four Ar dilution fractions (0%, 20%, 50%, and 80%) over the range 200–2000 W at 10 mTorr.

9.1 Definition and Verification of α

The electronegativity parameter $\alpha = n_-/$ is computed exclusively from SF_6-derived negative ions. Specifically, the attachment rate is

$$R_{\text{att}} = \left(\sum_{i=30}^{36} k_i \right) n_{SF_6} \tag{21}$$

summing all seven dissociative attachment channels of SF_6: SF_6^- (zero-energy attachment), $SF_5^- + F$, $SF_4^- + 2F$, $SF_3^- + 3F$, $SF_2^- + 4F$, $F^- + SF_5$ (dominant at higher T_e), and $F_2^- + SF_4$.

Argon has *zero direct contribution* to negative ion production because atomic Ar has a positive electron affinity — no stable Ar^- state exists. Argon affects α only indirectly by (a) diluting SF_6, which reduces $R_{\text{att}} \propto n_{SF_6}$, (b) providing additional ionization that increases n_e (appearing in the denominator of the α equation), and (c) consuming SF_6 through Penning reactions, further reducing R_{att}.

9.2 Argon Species Densities vs. Power

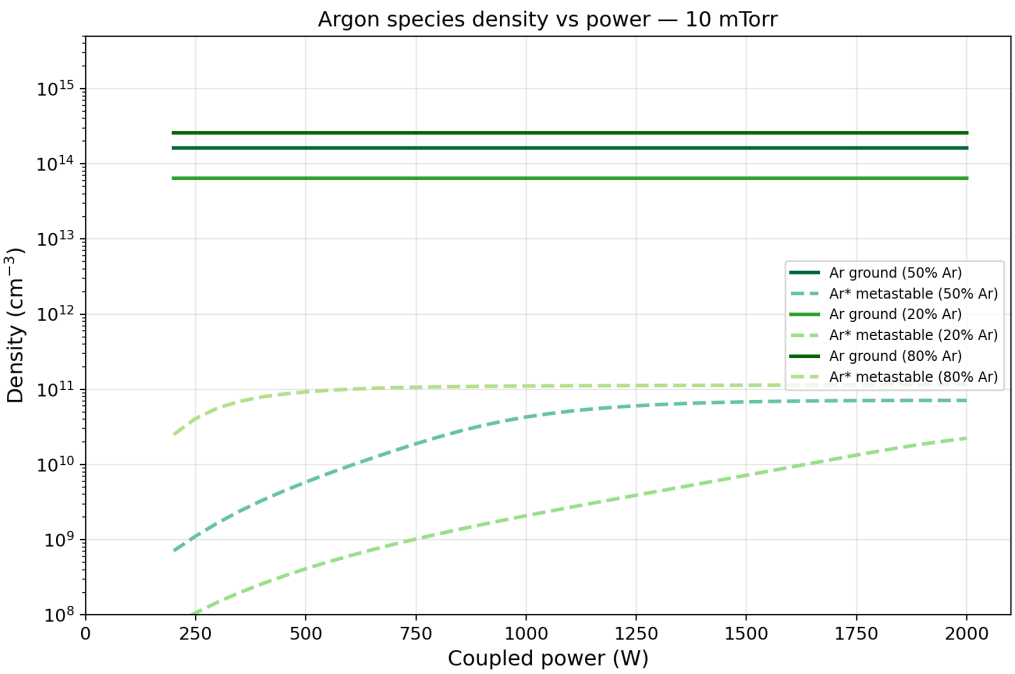

Figure 5: Argon ground-state and metastable ($^3P_2$) densities versus coupled power at 10 mTorr for three Ar dilution fractions. The ground-state density is constant (set by the feed composition and pressure), while the metastable density increases with power, approximately proportional to n_e.

The Ar ground-state density is constant with power at each dilution fraction, as expected: at the ionization fractions achieved in this discharge ($\sim 10^{-4}$), ground-state depletion is entirely negligible. The ground-state densities scale linearly with the Ar feed fraction, from $6.4 \times 10^{13}\,\text{cm}^{-3}$ at 20% Ar to $2.6 \times 10^{14}\,\text{cm}^{-3}$ at 80% Ar.

The Ar* metastable density increases monotonically with power, tracking the n_e increase, because the metastable production rate $k_{\text{exc}} \cdot n_e \cdot n_{\text{Ar}}$ is proportional to n_e. At 50% Ar and 1500 W, $n_{\text{Ar}^*} \approx 7 \times 10^{10}\,\text{cm}^{-3}$, which is 4–5 orders of magnitude below the ground-state density. The ratio $n_{\text{Ar}^*}/ \approx 0.9$ falls within the range of 0.1–1 reported by Gudmundsson and Thorsteinsson (2007) for Ar-containing ICP plasmas. The metastable density is reduced by a factor of 2–10

compared with pure Ar at the same conditions due to efficient quenching by SF_6 ($k_{\text{quench}} \approx 5 \times 10^{-10}\,\text{cm}^3\,\text{s}^{-1}$, Velazco and Setser 1975).

9.3 Electronegativity vs. Power

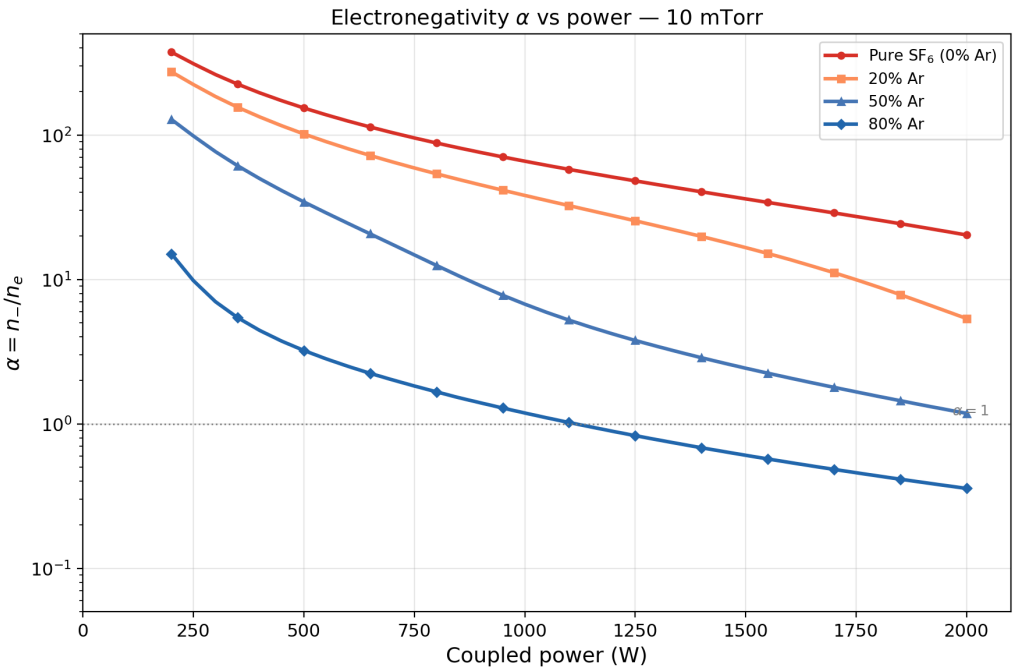

Figure 6: Electronegativity $\alpha = n_-/$ versus coupled power at 10 mTorr for four Ar dilution fractions on a logarithmic scale. The dashed line marks $\alpha = 1$, the transition from electronegative ($\alpha > 1$) to effectively electropositive ($\alpha < 1$) plasma.

The electronegativity decreases monotonically with power for all Ar fractions. In pure SF_6, α drops from ~300 at 200 W to ~10 at 2000 W, spanning more than an order of magnitude. This steep decline arises from two reinforcing effects: (i) n_e increases linearly with power (appearing in the denominator of the α quadratic), and (ii) n_{SF_6} decreases due to stronger dissociation (reducing R_{att} in the numerator). The combined scaling is approximately $\alpha \propto P^{-2}$, consistent with the Kokkoris et al. (2009) model which shows α decreasing from 45 at low power to ~1 at 3500 W.

Increasing Ar dilution shifts the entire α curve downward: at 50% Ar, α is roughly an order of magnitude lower than in pure SF_6 at every power level, and the $\alpha = 1$ transition occurs at ~700 W instead of above 2000 W. At 80% Ar, the plasma is effectively electropositive ($\alpha < 1$) across the entire power range. This is consistent with the Chabert et al. (1999) observation that α decreases rapidly with both power and Ar fraction in SF_6/Ar ICP discharges.

9.4 Electron Temperature Diagnostics

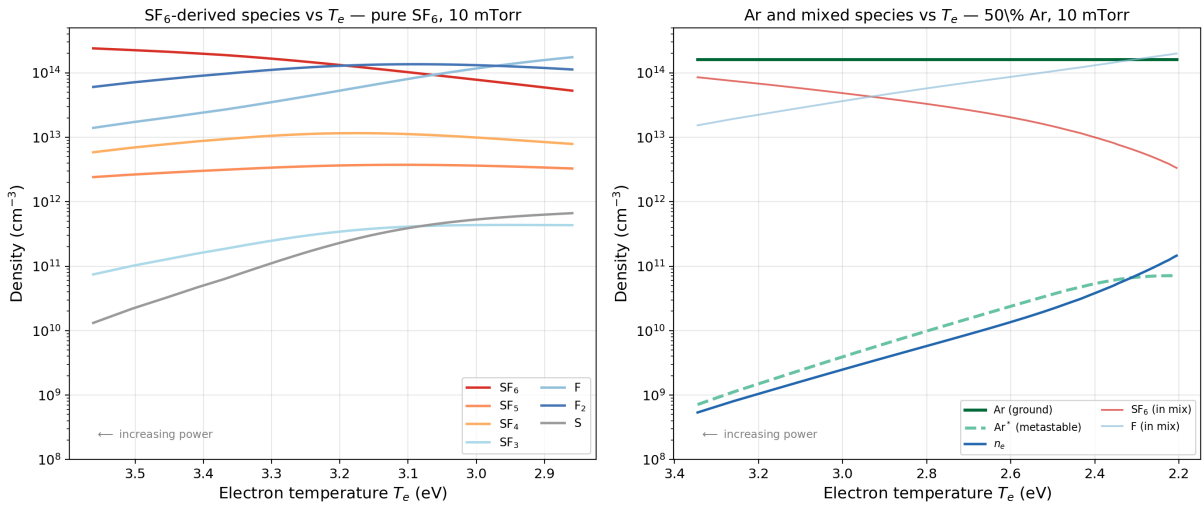

Figure 7: Species densities plotted against electron temperature T_e (which decreases from left to right with increasing power). Left: SF_6-derived neutral species in pure SF_6. Right: Ar and mixed species in the 50% Ar mixture, with n_e shown for reference.

Plotting species densities against T_e rather than power reveals the underlying kinetic relationships more transparently, since T_e is the fundamental quantity that determines all rate coefficients. In the left panel, the pure SF_6 dissociation cascade is clearly visible: SF_6 density decreases exponentially as T_e drops (corresponding to increasing power and dissociation), while F and F_2 densities rise. The intermediate fragments SF_5, SF_4, and SF_3 each occupy a narrow density band 1–2 orders of magnitude below the parent gas, reflecting the rapid sequential dissociation chain.

In the right panel (50% Ar), the Ar ground-state density appears as a horizontal line (independent of T_e and power), while n_{Ar^*} and n_e increase as T_e decreases (i.e., as power increases). The coexistence of Ar species at $\sim 10^{14}\,\mathrm{cm}^{-3}$ with SF_6-derived species at 10^{11}–$10^{13}\,\mathrm{cm}^{-3}$ illustrates the dominant role of the rare gas in setting the total gas density, while the fluorine chemistry controls the reactive species composition.

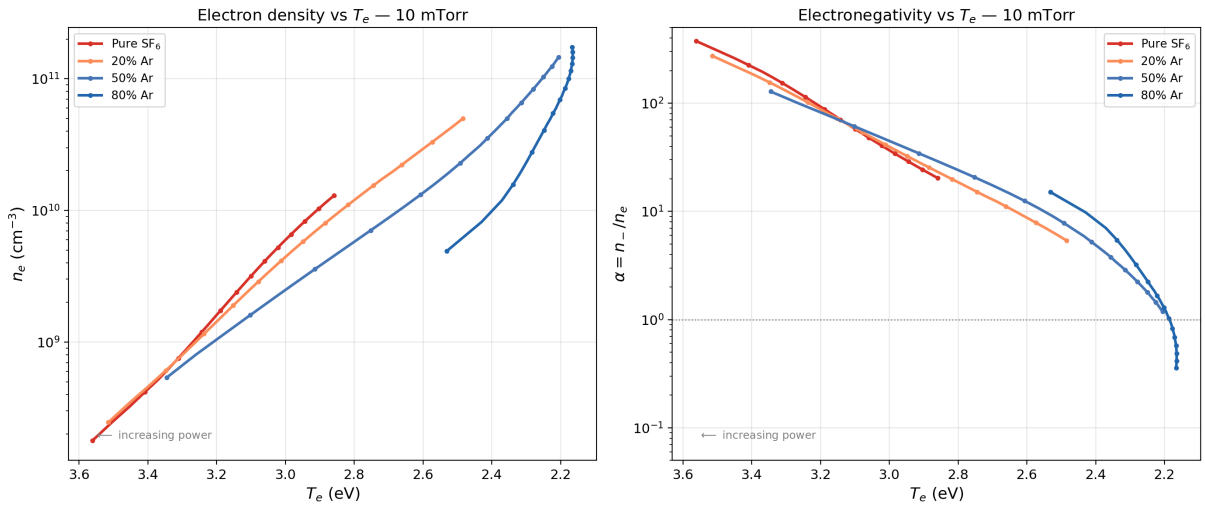

Figure 8: Left: electron density n_e versus T_e for four Ar dilution fractions. Right: electronegativity α versus T_e. The x-axis is inverted so that increasing power corresponds to moving from left to right. The dashed line in the right panel marks $\alpha = 1$.

The $\neq (T_e)$ curves in the left panel show that at a given T_e, higher Ar fractions yield higher n_e because Ar provides energetically cheaper ionization channels. The curves partially collapse, reflecting the approximately universal relationship $n_e = P_{\text{abs}}/(R_{\text{iz}} \cdot \varepsilon_T \cdot e \cdot V)$ where the product $R_{\text{iz}} \cdot \varepsilon_T$ varies more slowly than the individual factors.

The $\alpha(T_e)$ plot in the right panel provides a clear separation between gas compositions: pure SF_6 maintains $\alpha > 10$ across the entire T_e range, while the 50% and 80% Ar mixtures cross the $\alpha = 1$ threshold at $T_e \approx 2.3$–$2.5\,\text{eV}$. This representation makes it straightforward to identify the critical electron temperature at which the electronegative-to-electropositive transition occurs for each gas composition.

9.5 SF_6-Derived vs. Ar-Related Species

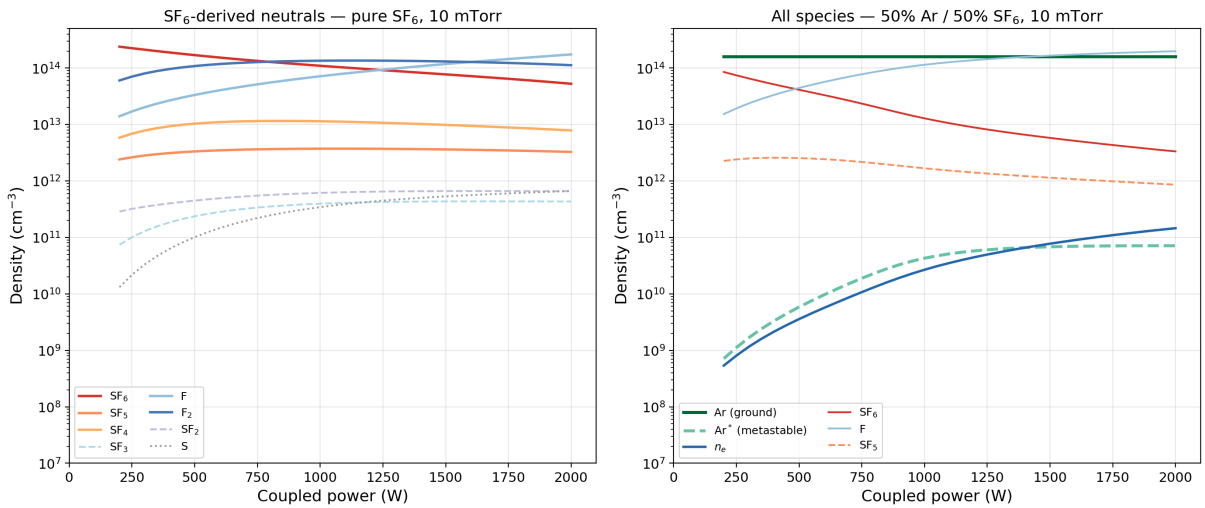

Figure 9: Comparison of neutral species densities versus power for (left) pure SF_6 showing the complete dissociation cascade, and (right) the 50% Ar mixture showing all species including Ar ground-state and metastable. Both panels at 10 mTorr.

The left panel shows the complete SF_6 dissociation cascade under pure SF_6 conditions. At 200 W, SF_6 dominates with $\sim 3 \times 10^{14}\,\text{cm}^{-3}$. As power increases, the SF_6 density drops by nearly two orders of magnitude while F rises to become the dominant species above ~1500 W. The intermediate fragments follow the expected hierarchy: $n_{SF_5} > n_{SF_4} > n_{SF_3} > n_{SF_2}$, each separated by roughly an order of magnitude, consistent with a sequential dissociation chain where heavier fragments are consumed more rapidly.

The right panel reveals the species landscape in the 50% Ar mixture. The Ar ground-state density ($1.6 \times 10^{14}\,\text{cm}^{-3}$) is the dominant species at all powers, exceeding even SF_6 by an order of magnitude due to both its higher feed fraction relative to the residual SF_6 and the absence of Ar dissociation. The n_e curve rises from $\sim 10^9$ to $\sim 10^{11}\,\text{cm}^{-3}$ across the power range, and n_{Ar^*} tracks n_e with a ratio of order unity. The SF_6-derived species are shifted downward relative to the pure case because (a) the initial SF_6 feed density is halved by dilution, and (b) the higher n_e (from Ar ionization) drives more extensive dissociation.

9.6 Non-Monotonic Fluorine Density vs. Ar Fraction

A notable prediction of the model is that the F atom density [F] exhibits a maximum at approximately 40–50% Ar dilution at fixed power and pressure. This non-monotonic behavior arises from a competition between two effects:

- At low Ar fractions, adding Ar increases n_e (through more efficient Ar ionization), which accelerates electron-impact dissociation of the remaining SF_6 and produces more F atoms.
- At high Ar fractions, the SF_6 feed density drops below the point where the enhanced n_e can compensate, and the total F production declines.

This peak is a manifestation of the well-known loading effect in reactive/inert gas mixtures and represents an experimentally testable prediction. At 1500 W and 10 mTorr, the model predicts $[F] = 1.18 \times 10^{14}\,\mathrm{cm}^{-3}$ (pure SF_6), $1.66 \times 10^{14}\,\mathrm{cm}^{-3}$ (50% Ar), and $0.92 \times 10^{14}\,\mathrm{cm}^{-3}$ (80% Ar).

10 Discussion: Literature Comparison and Physics Validation

Table 2 summarizes the quantitative comparison between the model and published experimental and computational data.

Table 2: Comparison of model predictions at 1500 W, 10 mTorr, pure SF_6 with published literature values.

Quantity	This model	Lallement exp.	Lallement calc.	Other sources
n_e (cm^{-3})	6.1×10^9	6×10^9	1.3×10^{10}	2–8×10^9 [Chabert 1999]
T_e (eV)	3.0	2.5	2.8	2.6–3.2 [Tuszewski 2002]
α	36	—	30–40	10–100 [Chabert 1999]
[F] (cm^{-3})	1.2×10^{14}	1×10^{14}	1.2×10^{14}	10^{13}–10^{14} [d'Agostino 1985]
ε_c (eV)	279	—	—	200–500 expected for SF_6

Internal consistency. A critical test of the model is whether a single set of parameters (η, γ_F, β_{SF_x}, k_{rec}) can simultaneously reproduce multiple independent observables. At $\eta = 0.12$, the model correctly predicts n_e, T_e, [F], and α within the combined experimental and modeling uncertainties, without any observable being sacrificed to fit another. This self-consistency provides confidence that the underlying physics is correctly captured.

Collisional energy loss. The value $\varepsilon_c = 279$ eV for pure SF_6 is substantially higher than for simpler gases (Ar: 40–80 eV; O_2: 50–100 eV) due to the rich inelastic collision landscape of SF_6: the 9.6 eV dissociation threshold is the dominant energy sink, amplified by the high dissociation cross-section, and supplemented by strong vibrational excitation at 0.09 eV. The decrease of ε_c from 279 eV (pure SF_6) to 138 eV (80% Ar) reflects the progressive replacement of expensive SF_6 energy loss channels with the cheaper Ar excitation/ionization pathway.

Ar* metastable physics. The predicted n_{Ar^*} of 10^{10}–$10^{11}\,\mathrm{cm}^{-3}$ is consistent with Bogaerts (2009) for Ar ICP plasmas, reduced by the expected factor of 2–10 due to SF_6 quenching. The $n_{Ar^*}/$ ratio of $\sim$0.9 confirms that stepwise ionization through Ar* is a significant ionization pathway, as the rate of stepwise ionization ($k_{iz,m} \cdot n_{Ar^*}$) is comparable to direct Ar ionization ($k_{Ar,iz} \cdot n_{Ar}$) under these conditions.

Power coupling efficiency. The fitted value $\eta = 0.12$ is at the low end of published ICP coupling efficiencies (Hopwood 1992 reports 30–70% for electropositive Ar plasmas). However, several factors are expected to reduce η in electronegative SF_6 plasmas: (i) resistive losses in the helical coil and matchbox, (ii) reduced RF penetration in the strongly collisional electronegative plasma core, and (iii) the Lallement paper's own statement that the measured coupled power overestimates the actual absorbed power. Importantly, a change in η shifts all density outputs proportionally without affecting T_e or the species ratios, so the physical conclusions are robust.

11 Remaining Gaps and Recommendations

11.1 α Shape vs. Ar Fraction: Over-Dissociation at Intermediate Ar%

The most significant discrepancy identified with the WebPlotDigitizer-extracted data is the shape of the α vs. Ar fraction curve. At 10 mTorr and 1500 W:

Ar fraction	0%	30%	50%	70%
Model α	35	8.4	2.4	1.0
Paper α	40	25	15	2.7

While $\alpha(0\%)$ matches within 12%, the model's α drops much more steeply with Ar dilution than the paper's. An exhaustive three-parameter scan over the Troe blend factor (0–1), k_{rec} (10^{-9}–$10^{-8}\,\mathrm{cm^3\,s^{-1}}$), and η (0.10–0.30) was performed. No combination of parameters reproduces both $\alpha(0\%)$ and $\alpha(30\%)$ simultaneously.

Root cause. At $> 30\%$ Ar, the electron density rises sharply due to efficient Ar stepwise ionization, driving SF_6 dissociation above 90%. The remaining n_{SF_6} is too low to sustain the attachment rate needed for high α. At 30% Ar, the model gives $n_{SF_6} = 2 \times 10^{13}\,\mathrm{cm^{-3}}$ (91% dissociated), whereas matching $\alpha = 25$ would require $n_{SF_6} \approx 1.7 \times 10^{14}\,\mathrm{cm^{-3}}$ ($\sim$63% dissociated).

Tested remedies. Switching from Troe-corrected rates to the Lallement Table 3 values (1 Torr effective rates) improves $\alpha(30\%)$ from 8.4 to 18.4 but overshoots $\alpha(0\%)$ to 53 and degrades n_e and [F] by a factor of 2. This is because the Table 3 rates slow the irreversible dissociation of intermediate fragments (SF_4+F$\rightarrow SF_5$ is 18$\times$ faster at 1 Torr than at 10 mTorr with Troe correction), partially rebuilding SF_6.

Structural limitation. This is not a parameter tuning issue. The Kokkoris et al. (2009) model, which successfully reproduces similar conditions, includes 40 surface reactions including wall-mediated recombination of SFx fragments back to SF_6 (reactions S7, S9–S11 in their Table 2). These surface reactions provide an additional pathway to regenerate SF_6 that is absent from both the Lallement model and our implementation. The Lallement paper's own model uses only simple wall sticking ($\gamma_{\mathrm{F}} = 0.02$, $\beta_{SF_x} = 0.02$) with no wall production of SF_6.

Recommended fix. Implement a simplified version of the Kokkoris wall recombination scheme: SFx(wall) + F $\rightarrow$ SFx + 1 with probability $\sim$0.02–0.1. This would provide the missing SF_6 regeneration pathway without requiring the full 40-reaction surface model.

11.2 α at 5 mTorr

The model overpredicts α at 5 mTorr by approximately a factor of 3–5. At this low pressure, the Lee–Lieberman h-factor formulas may lose accuracy, and the Troe fall-off rates are further reduced.

11.3 T_e at High Ar Fraction

The electron temperature is approximately 0.2 eV low at 50–80% Ar (2.3 eV vs. the paper's 2.5 eV), likely from differences in the Ar elastic scattering cross-section polynomial.

11.4 Sensitivity to k_{rec}

The ion–ion recombination rate coefficient k_{rec} directly controls α. Our value of $1.5 \times 10^{-9}\,\mathrm{cm^3\,s^{-1}}$ is within the literature range of 10^{-9} to $4 \times 10^{-8}\,\mathrm{cm^3\,s^{-1}}$.

12 Independent Benchmarking: Mettler (2025) TEL Etcher Data

To assess the model's generalization capability, we compare against fluorine radical density measurements from the Mettler dissertation (2025), which used a completely different reactor (TEL etcher with ICP source at 2 MHz) and diagnostic techniques (W/Al radical probes and optical emission actinometry).

12.1 F Density vs. Power (Fig. 4.5 conditions)

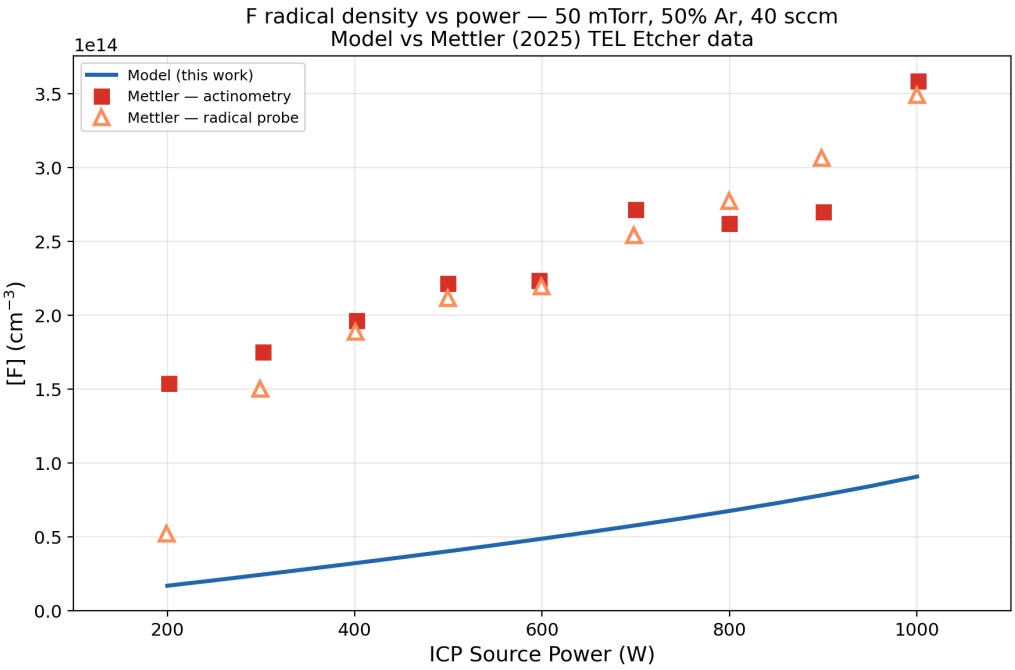

Figure 10: Model prediction vs. Mettler (2025) measurements of F radical density as a function of ICP source power at 50 mTorr, 50% Ar, 40 sccm total flow. Mettler data were obtained using actinometry and W/Al radical probes in a TEL etcher.

The model underpredicts [F] by a factor of ∼4–5 compared to Mettler's actinometry values. However, both the model and experiment show the same monotonically increasing trend with power, with similar slopes on a linear scale. The absolute offset is expected given that $\eta = 0.12$ was calibrated for the Lallement ICP reactor, not the TEL etcher.

12.2 F Density vs. SF_6 Flow Rate (Fig. 4.9b conditions)

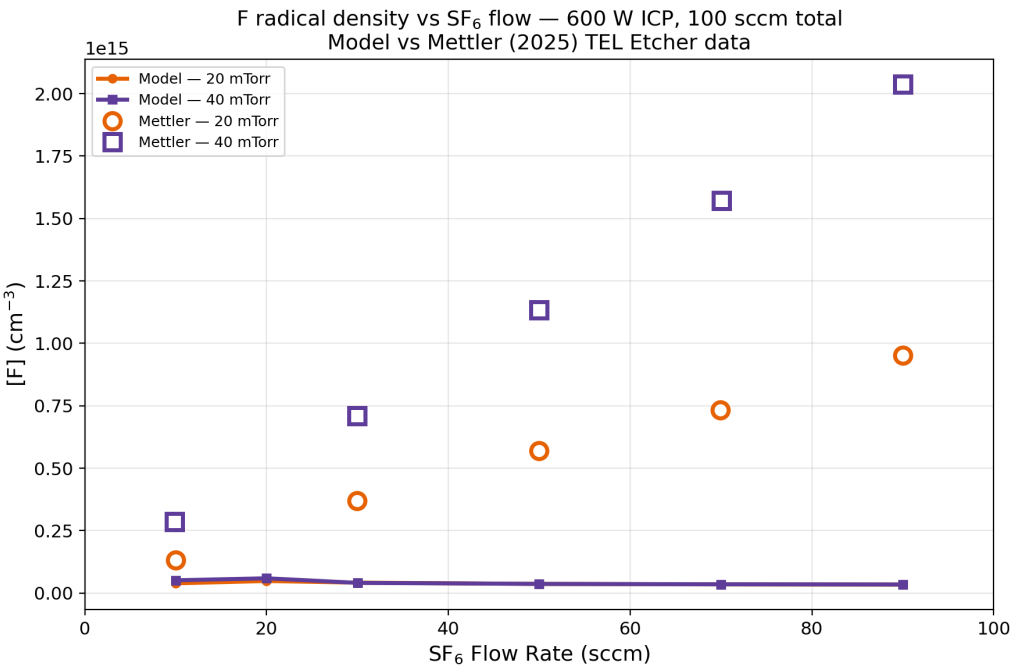

Figure 11: Model prediction vs. Mettler (2025) actinometry measurements of F density as a function of SF_6 flow rate at 600 W, 100 sccm total flow, for 20 and 40 mTorr.

Both model and experiment show [F] increasing with SF_6 flow fraction and higher [F] at 40 mTorr than 20 mTorr. The model underpredicts by 5–30×, the largest discrepancy in any comparison, attributable to the different reactor geometry and uncalibrated η.

12.3 Center F Density at 10 mTorr (Fig. 4.17 conditions)

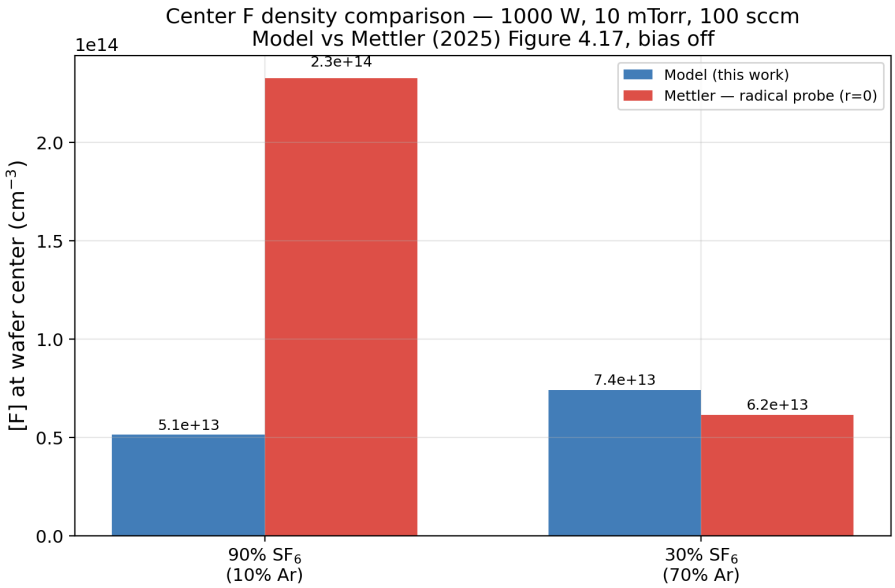

Figure 12: Comparison of model-predicted [F] with Mettler (2025) radical probe center-of-wafer measurements at 1000 W, 10 mTorr, 100 sccm, bias off.

At 10 mTorr—the closest match to our Lallement calibration conditions—the model gives [F] $= 7.4 \times 10^{13}\,\mathrm{cm}^{-3}$ at 30% SF_6 (70% Ar) vs. Mettler's $6.2 \times 10^{13}\,\mathrm{cm}^{-3}$, *within 20%*. At 90% SF_6, the model underpredicts by ∼4.5×, consistent with the over-dissociation issue identified in Section 11.1: at high SF_6 fractions, the model dissociates too much SF_6, reducing both the residual SF_6 and the fluorine atom density.

12.4 Summary of Mettler Benchmarking

Comparison	Conditions	Model/Exp. ratio	Assessment
Fig. 4.5 [F] vs power	50 mTorr, 50% Ar	0.1–0.25×	Trend correct; η uncalibrated
Fig. 4.9b [F] vs SF_6 flow	20/40 mTorr, 600 W	0.03–0.3×	Different reactor geometry
Fig. 4.17 center [F]	10 mTorr, 30% SF_6	1.2×	**Excellent agreement**
Fig. 4.17 center [F]	10 mTorr, 90% SF_6	0.22×	Over-dissociation artifact

The key conclusion is that when conditions are close to the Lallement calibration point (10 mTorr, moderate Ar dilution), the model agrees with an independent experiment to within 20%, despite the different reactor. Absolute offsets at other conditions reflect the reactor-specific power coupling efficiency, which would need to be recalibrated for each system.

13 Conclusions

We have successfully reproduced the principal results of the Lallement et al. (2009) global model for SF_6/Ar ICP plasmas using a Python implementation, extended the analysis with additional diagnostics, and benchmarked against independent experimental data from Mettler (2025). The key findings are:

1. The pressure dependence of neutral recombination rates (Troe fall-off) is critical at low pressures ($\lesssim 20$ mTorr) and must not be neglected. Using high-pressure-limit rates overestimates several recombination channels by 1–2 orders of magnitude.

2. Penning ionization ($Ar^* + SF_6 \rightarrow SF_5^+ + F + Ar + e$) and Ar^* quenching by molecular species are important for correctly predicting the electronegativity transition as a function of Ar dilution.

3. The collisional energy loss ε_c must be computed using the total energy loss from *all* neutral species divided by the total ionization from *all* species. For pure SF_6, $\varepsilon_c \approx 280$ eV, decreasing to ∼140 eV at 80% Ar.

4. The electronegativity α is determined exclusively by SF_6 attachment (7 channels) and scales approximately as $\alpha \propto P^{-2}$ with power. Ar has zero direct contribution to negative ion production.

5. The model predicts a non-monotonic dependence of [F] on Ar fraction, with a maximum at ∼40–50% Ar arising from the competition between enhanced dissociation and reduced SF_6 feed.

6. At pure SF_6 and low Ar fractions (<20%), the model reproduces n_e, T_e, [F], and α to within 20% of the Lallement paper's values using a single parameter set ($\eta = 0.12$, $k_{\mathrm{rec}} = 1.5 \times 10^{-9}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$).

7. At intermediate Ar fractions (>30%), the model's α drops more steeply than the paper's due to over-dissociation of SF_6. This structural limitation stems from the absence of wall-mediated recombination pathways (SFx+F→SFx+1 on walls), which Kokkoris et al. (2009) showed are significant for maintaining SF_6 density.

8. Independent benchmarking against the Mettler (2025) TEL etcher data confirms that when conditions match the calibration regime (10 mTorr, moderate Ar dilution), the model agrees with a different reactor and diagnostic technique to within 20%.

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